Addition in $\mathbb{Q}$

1. $\frac{2}{7} + \frac{3}{7}$
2. $\frac{1}{4} + \frac{3}{8}$
3. $\frac{5}{6} + \frac{7}{9}$
4. $-\frac{3}{5} + \frac{7}{10}$
5. $2\frac{1}{3} + 1\frac{3}{4}$
6. $-\frac{5}{8} + \left(-\frac{1}{6}\right)$
7. $\frac{3}{4} + \left(-\frac{7}{12}\right) + \frac{5}{6}$
8. $-3\frac{2}{5} + 5\frac{1}{2}$
9. A recipe calls for $\frac{3}{4}$ cup of white sugar and $\frac{2}{3}$ cup of brown sugar. What is the total amount of sugar required for the recipe?
10. A hiker walked $3\frac{1}{2}$ miles in the morning and $2\frac{4}{5}$ miles in the afternoon. What was the total distance walked by the hiker?

Solutions

1. $\frac{5}{7}$

   Explanation: The fractions share a common denominator of 7. Add the numerators directly:

   $$\frac{2}{7} + \frac{3}{7} = \frac{2+3}{7} = \frac{5}{7}.$$

2. $\frac{5}{8}$

   Explanation: Find a common denominator ($\text{LCM}(4, 8) = 8$). Convert $\frac{1}{4}$ to an equivalent fraction with denominator 8:

   $$\frac{1}{4} = \frac{1 \times 2}{4 \times 2} = \frac{2}{8}.$$

   Add the numerators:

   $$\frac{2}{8} + \frac{3}{8} = \frac{2+3}{8} = \frac{5}{8}.$$

3. $\frac{29}{18}$ **or** $1\frac{11}{18}$

   Explanation: Find a common denominator ($\text{LCM}(6, 9) = 18$). Convert both fractions:

   $$\frac{5}{6} = \frac{5 \times 3}{6 \times 3} = \frac{15}{18}, \quad \frac{7}{9} = \frac{7 \times 2}{9 \times 2} = \frac{14}{18}.$$

   Add the numerators:

   $$\frac{15}{18} + \frac{14}{18} = \frac{29}{18} = 1\frac{11}{18}.$$

4. $\frac{1}{10}$

   Explanation: Find a common denominator ($\text{LCM}(5, 10) = 10$). Convert $-\frac{3}{5}$:

   $$-\frac{3}{5} = -\frac{3 \times 2}{5 \times 2} = -\frac{6}{10}.$$

   Combine the signed numerators:

   $$-\frac{6}{10} + \frac{7}{10} = \frac{-6+7}{10} = \frac{1}{10}.$$

5. $\frac{49}{12}$ **or** $4\frac{1}{12}$

   Explanation: Convert mixed numbers to improper fractions:

   $$2\frac{1}{3} = \frac{7}{3}, \quad 1\frac{3}{4} = \frac{7}{4}.$$

Find a common denominator ($\text{LCM}(3,4) = 12$):

$$\frac{7}{3} = \frac{28}{12}, \quad \frac{7}{4} = \frac{21}{12}.$$

Add:

$$\frac{28}{12} + \frac{21}{12} = \frac{49}{12} = 4\frac{1}{12}.$$

6. $-\dfrac{19}{24}$

Explanation: Find a common denominator ($\text{LCM}(8,6) = 24$). Convert fractions:

$$-\frac{5}{8} = -\frac{15}{24}, \quad -\frac{1}{6} = -\frac{4}{24}.$$

Combine negative numerators:

$$-\frac{15}{24} + \left(-\frac{4}{24}\right) = \frac{-15-4}{24} = -\frac{19}{24}.$$

7. 1

Explanation: Find a common denominator ($\text{LCM}(4,12,6) = 12$). Convert terms:

$$\frac{3}{4} = \frac{9}{12}, \quad -\frac{7}{12} = -\frac{7}{12}, \quad \frac{5}{6} = \frac{10}{12}.$$

Combine numerators:

$$\frac{9-7+10}{12} = \frac{12}{12} = 1.$$

8. $\dfrac{21}{10}$ **or** $2\dfrac{1}{10}$

Explanation: Convert to improper fractions:

$$-3\frac{2}{5} = -\frac{17}{5}, \quad 5\frac{1}{2} = \frac{11}{2}.$$

Find a common denominator of 10:

$$-\frac{17}{5} = -\frac{34}{10}, \quad \frac{11}{2} = \frac{55}{10}.$$

Add:

$$-\frac{34}{10} + \frac{55}{10} = \frac{-34+55}{10} = \frac{21}{10} = 2\frac{1}{10}.$$

9. $\dfrac{17}{12}$ **cups or** $1\dfrac{5}{12}$ **cups**

Explanation: Add $\dfrac{3}{4}$ and $\dfrac{2}{3}$. Common denominator is 12:

$$\frac{3}{4} = \frac{9}{12}, \quad \frac{2}{3} = \frac{8}{12}.$$

Sum the quantities:

$$\frac{9}{12} + \frac{8}{12} = \frac{17}{12} = 1\frac{5}{12} \text{ cups}.$$

10. $\frac{63}{10}$ **miles or** $6\frac{3}{10}$ **miles**

Explanation: Convert $3\frac{1}{2} = \frac{7}{2}$ and $2\frac{4}{5} = \frac{14}{5}$. Common denominator is 10:

$$\frac{7}{2} = \frac{35}{10}, \quad \frac{14}{5} = \frac{28}{10}.$$

Sum the distances:

$$\frac{35}{10} + \frac{28}{10} = \frac{63}{10} = 6\frac{3}{10} \text{ miles.}$$